Invention Title:
Ergonomic Movie Theater Configuration

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Field of the Invention:
The present invention relates generally to theaters, and in particular, to movie theaters.

Background of the Invention:

Traditional Movie Theaters

The first movie theaters were mostly converted from classic stage theaters more than 100 years ago. Since then, traditional movie theater configuration has not changed much. This configuration includes a screen vertically positioned on a stage and an audience area with a stepped floor sloped down towards the screen. The audience area includes a plurality of seats facing the screen and arranged side by side into the rows. The rows, on the other hand, are arranged one after another, each on the different height of the stepped floor.

This traditional design became widely adopted, and because of that almost all movie theaters in the world have three serious deficiencies – distraction, discomfort, and inequality:

– *Distraction.* Since the screen is positioned in front of the audience and the slope of the audience area is relatively low, the patrons occupying the front rows may somehow distract other patrons sitting at the back.

– *Discomfort.* Since the screen is positioned vertically in front of the audience, the theater patrons must keep their bodies and heads in the upright position for a long time. The research carried out using magnetic resonance imaging (MRI) at the Centre for Spinal Research in Aberdeen, revealed that the upright sitting posture causes increased pressure on the lumbar spine, spinal disks, and associated muscles and tendons, leading to the feelings of discomfort and pain in neck muscles and lower back muscles during a prolonged sitting.

– *Inequality.* The front rows are too close to the screen, while the rear rows are too distant from the screen, and only a few rows are considered as the best among the theater patrons.

Present Day Movie Theaters

With the improvements in image projection technology, present day movie theater configuration adopts three common solutions to the deficiencies of traditional design – an increased slope of the audience area, a larger screen, and higher, reclining seat backs. However, although widely adopted, these solutions have their limitations:

– Since the screen is still positioned in front of the audience, some patrons occupying the front rows may still somehow distract other patrons sitting at the back, even with higher seat backs.

– Since the screen is still positioned vertically in front of the audience, the theater patrons should keep their heads almost upright watching a movie. During a prolonged sitting, this posture induces discomfort, no matter how comfortable the seats are.

– With the increased slope of the audience area, there may be fewer rows or less space for each row. Additionally, the legs of the theater patrons may be on the same level as the heads of other patrons sitting in the front row. This causes even more discomfort.

– Since the distance from the audience to the screen has not changed, with a larger screen, there may be even more inequality between the rows.

Summary of the Invention:

A purpose of the present invention is to provide a novel, ergonomic movie theater configuration which eliminates or minimizes the major deficiencies of prior art movie theaters.

This configuration adopts one simple solution – a change in the angle and position of the screen, not the slope of the audience area. In other words, in ergonomic movie theater configuration, the screen is inclined towards the audience area and positioned higher relative to prior art movie theaters, preferably above the audience area.

This solution provides three positive outcomes:

– *No distraction.* In ergonomic movie theater configuration, the screen is raised at the optimal level and inclined at the optimal degree, so as the patrons occupying the front rows simply do not enter the viewing area of other patrons sitting in the back.

– *No discomfort.* The forenamed research revealed that the optimal sitting position was reclining back at approximately 135 degrees. This sitting position was shown to cause the least strain on the lumbar spine, spinal disks, and associated muscles and tendons. In ergonomic movie theater configuration, all of the seats are reclined at the optimal degrees, so as to ensure the most comfortable position to all patrons.

– *More equality.* Since the screen is inclined towards the audience, the top edge of the screen is closer to the rear rows, while the bottom edge is more distant from the front rows, relative to prior art movie theaters. This minimizes the difference in viewing distances and viewing angles among the patrons occupying different rows, and ensures that each patrons has the best viewing experience possible.

Brief Description of the Drawings:

It is to be appreciated that the foregoing paragraphs have been provided by way of general introduction, and are not intended to limit the spirit and scope of the present invention. The various exemplary embodiments (and/or elements of the exemplary embodiments) of the present invention, together with further advantages, will be best appreciated by reference to the following detailed description taken in conjunction with the accompanying drawings:

FIG. 1 is an elevation view, sectioned by a vertical plane, of an exemplary embodiment of traditional movie theater configuration. This figure is labeled ***PRIOR ART***.

FIG. 2 is an elevation view, sectioned by a vertical plane, of an exemplary embodiment incorporating three middle rows of the traditional movie theater. This figure is labeled ***PRIOR ART***.

FIG. 3 is an elevation view, sectioned by a vertical plane, of an exemplary embodiment of present day movie theater configuration. This figure is labeled ***PRIOR ART***.

FIG. 4 is an elevation view, sectioned by a vertical plane, of an exemplary embodiment incorporating three middle rows of the present day movie theater. This figure is labeled ***PRIOR ART***.

FIG. 5 is an elevation view, sectioned by a vertical plane, of an exemplary embodiment of ergonomic movie theater configuration.

FIG. 6 is an elevation view, sectioned by a vertical plane, of an exemplary embodiment incorporating three middle rows of the ergonomic movie theater.

FIG. 7A, ***FIG. 7B***, and ***FIG. 7C*** are the elevation views, sectioned by a vertical plane, of the exemplary embodiments incorporating the foremost row, the middle row, and the rearmost row of the ergonomic movie theater respectively.

FIG. 8A and ***FIG. 8B*** are the elevation views, sectioned by a vertical plane, of various exemplary embodiments of the auditoriums incorporating ergonomic movie theater configuration.

Detailed Description of the Invention:

Referring now to the drawings wherein like reference letters correspond to similar elements throughout the several views and more specifically, referring to ***FIG. 1***, ***FIG. 3***, and ***FIG. 5***, the present invention will be described in the context of various exemplary movie theater configurations which include four interrelated parameters:

- a slope ***a'*** of an audience area ***A***,
- an angle ***s'*** of inclination of a screen ***S*** with respect to horizontal axis ***x***,
- a vertical distance ***D*** from the lowest level of the audience area ***A*** to the bottom edge ***L*** of the screen ***S***, and
- a horizontal distance ***W*** from the lowest level of the audience area ***A*** to the bottom edge ***L*** of the screen ***S***.

Additionally, in ***FIG. 1***, ***FIG. 3***, and ***FIG. 5***, a point ***P*** represents an approximate position of a projection lamp and a point ***C*** represents a center of the screen. The points ***U*** and ***L*** represent a top edge and a bottom edge of the screen respectively. The broken lines represent the total vertical viewing angles of the screen for the eyes of patrons occupying each row.

Referring now to ***FIG. 1***, it is to be appreciated that in traditional movie theater configuration, the slope ***a'*** is relatively low, usually between 0 and +1/5.

This is due to corresponding image projection technology used. Traditional configuration was adopted in the times when there were no methods of image geometry correction. Thus, there was a need to install a projection apparatus directly opposite the center of the screen, behind the rearmost row.

Subsequently, with the improvements in image projection technology, especially with the introduction of the methods of image geometry correction, it became possible to install the projection apparatus higher than the center of the screen and increase the slope of the audience area.

Thus, referring to ***FIG. 3***, it is to be appreciated that in present day movie theater configuration, the slope ***a'*** is relatively higher with respect to traditional configuration, and usually between +1/5 and +1.

Referring now to ***FIG. 5***, it is to be appreciated that in ergonomic movie theater configuration, the slope ***a'*** is again relatively low, and usually between 0 and +1/2.

Referring again to ***FIG. 1***, it is to be appreciated that in traditional movie theater configuration, the angle ***s'*** is usually upright, or equals 90 degrees.

Subsequently, with the possibility to install the projection apparatus higher than the center of the screen, and increase the slope of the audience area, there was a need to recline the screen slightly outwards the audience area, to provide a better view for all patrons.

Thus, referring to ***FIG. 3***, it is to be appreciated that in present day movie theater configuration, the angle ***s'*** is slightly more than 90 degrees, and usually between 90 and 100 degrees.

Referring now to ***FIG. 5***, it is to be appreciated that in ergonomic movie theater configuration, the screen is inclined towards the audience area. Thus, in ergonomic configuration, the top edge ***U*** of the screen is relatively closer to the audience area, while the bottom edge ***L*** is relatively more distant from the audience area, with respect to traditional and present day configurations. This is to minimize the difference in viewing distances and viewing angles among the patrons occupying different rows and to ensure that ergonomic configuration provides the best viewing experience to all patrons.

Thus, in the ergonomic configuration, the angle ***s'*** is less than 90 degrees, usually between 40 and 80 degrees, and preferably equals 60 degrees.

Referring again to ***FIG. 1***, it is to be appreciated that in traditional movie theater configuration, the distance ***D*** is relatively low, usually between 0 and 1 meter.

Referring to ***FIG. 3***, it is to be appreciated that in present day movie theater configuration, the distance ***D*** is usually similar to that in the traditional configuration.

However, with the possibility to increase the slope of the audience area, there was a need to increase the size of the screen, to provide a better view for all patrons. Thus, in the present day movie theaters, the screen is relatively larger with respect to the traditional movie theaters.

Referring now to ***FIG. 5***, it is to be appreciated that in ergonomic movie theater configuration, the distance ***D*** is relatively higher with respect to traditional and present day configurations, usually more than 1 meter, and preferably the screen is positioned above the audience area.

Referring again to ***FIG. 1***, ***FIG. 3***, and ***FIG. 5***, the distance ***W*** is relatively the same with respect to all configurations.

In summary, the main features of various movie theater configurations are provided below:

- *traditional:*
  - ***a'*** ≈ from 0 to +1/5;
  - ***s'*** = 90°, or ≈ 90°;
  - ***D*** < 1m;
  - ***W –*** no change;
- *present day:*
  - ***a'*** ≈ from +1/5 to +1;
  - ***s'*** > 90°, or ≈ 90-100°;
  - ***D*** < 1m;
  - ***W –*** no change;
- *ergonomic:*
  - ***a'*** ≈ from 0 to +1/2;
  - ***s'*** < 90°, or ≈ 40-80°;
  - ***D*** > 1m;
  - ***W –*** no change.

Referring now to ***FIG. 2***, ***FIG. 4***, ***FIG. 6***, ***FIG. 7A***, ***FIG. 7B***, and ***FIG. 7C***, a point ***E*** represents a position of the eyes of the theater patron occupying a corresponding row ***M***. A line ***EC*** represents a direction of a patron's eyesight towards the center ***C*** of the screen. The line ***EC*** also represents a line of straight eyesight of the patron's eyes. The lines ***EU*** and ***EL*** represent the directions of the patron's eyesight toward a top edge ***U*** and a bottom edge ***L*** of the screen respectively. Thus, an angle ***UEL*** represents a total vertical viewing angle of the screen for the patron's eyes. An angle ***v'*** represents an angle of a maximum eye movement of the patron's eyes, which incorporates approximately 25 degrees of an upward eye movement and approximately 30 degrees of a downward eye movement with respect to the line of straight eyesight (or the line ***EC***).

Additionally, in ***FIG. 2***, ***FIG. 4***, and ***FIG. 6***, a height ***B*** represents a height of a seat back of a usual theater seat. The distances ***M*** and ***F*** represent two different rows arranged one after another.

Referring now to ***FIG. 2***, it is to be appreciated that in traditional movie theaters, the height ***B*** is relatively low, so as the seat back is usually lower than a head of an average theater patron occupying the corresponding seat. Given that in traditional configuration the slope of the audience area is relatively low, the heads of the patrons occupying the front row ***F*** may partly enter the viewing angle ***UEL*** of the patrons occupying the row ***M***. This causes distraction among the theater patrons toward one another.

Referring to ***FIG. 4***, it is to be appreciated that in present day movie theaters, the height ***B*** is relatively higher with respect to traditional theaters, so as the seat back is usually higher than the head of the average theater patron occupying the corresponding seat. Given that in present day movie theater configuration the slope of the audience area is relatively high, the heads of the patrons occupying the front row ***F*** usually do not enter the viewing angle ***UEL*** of the patrons occupying the row ***M***. However, the seat backs of the seats of the front row ***F*** partly enter the angle ***v'*** of the maximum eye movement of the patrons occupying the row ***M***. This may cause distraction among the theater patrons toward one another.

Referring now to ***FIG. 6***, it is to be appreciated that in ergonomic movie theaters, the height ***B*** is relatively moderate with respect to traditional and present day theaters, so as the seat back usually ends at the same level as the head of the average theater patron occupying the corresponding seat. In ergonomic movie theater configuration, the interrelated parameters ***a'***, ***s'***, and ***D*** must be selected in such a way, so as neither the heads of the patrons occupying the front rows ***F*** nor the seat backs of the seats of the front row ***F*** enter into the angle ***v'*** of the maximum eye movement of the patrons occupying the row ***M***. This means that in ergonomic movie theater configuration, there must be no distraction among the theater patrons toward one another.

Referring now to ***FIG. 7A***, ***FIG. 7B***, and ***FIG. 7C***, a distance ***R*** represents a space required for a corresponding row. A line ***H*** represents a vertical axis of the patron's head. This line is perpendicular to the line of straight eyesight of the patron's eyes (or the line ***EC***). An angle ***h'*** represents an angle of a maximum declination of the patron's head (or the line ***H***) with respect to vertical axis ***y***.

Referring to ***FIG. 7A***, in the case of the foremost row of the ergonomic movie theater, the angle ***h'*** is relatively high. This is due to the fact that in this row, the seat backs of the seats must be reclined more, so as the patrons occupying this row look exactly at the center of the screen. Additionally, in this case, the distance ***R*** is also relatively high. This is due to the fact there must be more space for this row.

Referring to ***FIG. 7B***, in the case of the middle row of the ergonomic movie theater, the angle ***h'*** is relatively lower with respect to the foremost row. This is due to the fact that in this row, the seat backs of the seats must be reclined less with respect to the foremost row. Additionally, in this row, the distance ***R*** is also relatively lower with respect to the foremost row. This is due to the fact that there must be less space for this row.

Referring to ***FIG. 7C***, in the case of the rearmost row of the ergonomic movie theater, the angle ***h'*** is relatively lower with respect to the middle row. This is due to the fact that in this row, the seat backs of the seats must be reclined even less with respect to the foremost row. Additionally, in this case, the distance ***R*** is also relatively lower with respect to the middle row. This is due to the fact that there must be even less space for this row.

In summary, in the ergonomic movie theater, the angle ***h'*** and the distance ***R*** must gradually decrease from the foremost row to the rearmost row. This is to ensure that all of the patrons in each row look exactly at the center of the screen and that ergonomic configuration provides the best viewing experience to all patrons.

Additionally, it is to be appreciated that the angle ***h'*** also represents an angle of a maximum declination of the corresponding seat back. This is due to the fact that in the ergonomic movie theater, each patron's back and head rest completely on the corresponding seat back. Thus, there must be no feelings of discomfort, neck pain, and back pain among the theater patrons.

In summary, the various exemplary embodiments (and/or elements of the exemplary embodiments) disclosed herein are believed to result in a movie theater configuration that eliminates or minimizes the three major deficiencies of prior art movie theater configurations – distraction, discomfort, and inequality. Thus, the present invention is well adapted to carry out its objectives and attain the ends and advantages mentioned above as well as those inherent therein.

It is to be appreciated that the foregoing detailed description of the exemplary embodiments (and/or elements of the exemplary embodiments) has been provided for the purpose of explaining the general principles and practical application of the present invention, thereby enabling others skilled in the art to understand the present invention for various embodiments and with various modifications as are suited to the particular use contemplated. This specification is not necessarily intended to be exhaustive or to limit the present invention to the exemplary embodiments (and/or elements of the exemplary embodiments) disclosed. Any of the embodiments (and/or elements of the exemplary embodiments) disclosed herein may be altered and/or modified and/or combined with one another to form various additional embodiments not specifically disclosed. Accordingly, additional embodiments are possible and are intended to be encompassed within this specification. This specification describes specific examples to accomplish a more general goal that may be accomplished in another way. Thus, this specification is to cover all alternatives, combinations, equivalents, and modifications, falling within the spirit and scope of the present invention.

It is to be appreciated that while the exemplary embodiments (and/or elements of the exemplary embodiments) of the present invention have been described herein with respect to a theater specifically designated for movie distribution, the same embodiments are equally applicable to a theater distributing any other type of visual media (advertisements, concerts, drawings, images, paintings, photography, presentations, video art, video clips, video games, video streams, websites, etc.). Additionally, it is to be appreciated that the present invention may be incorporated into new theaters or as features added during a retrofit of an existing theater.

A theater may be either an indoor theater or an outdoor theater, or any other type (underground, underwater, etc.).

The theater may have any number of auditoriums. An elevation view of each auditorium in the theater may be either in a form of a rectangle or a dome, or any other form. All of the auditoriums in the theater may have the same form or different forms depending on individual design. ***FIG. 8A*** and ***FIG. 8B*** illustrate the elevation views, sectioned by a vertical plane, of various exemplary embodiments of the auditoriums incorporating ergonomic movie theater configuration.

The theater may have any number of audience areas. Each audience area in the theater may have either a constant slope or an altering slope, or any other type. All of the audience areas in the theater may have the same slope or different slopes depending on individual design.

Each audience area in the theater may have any number of rows. Each row in the audience area may be either a straight row or a curved row (e.g. arc-shaped), or any other type. All of the rows in the audience area may be of the same type or of the different types depending on individual design.

Each audience area in the theater may have any number of seats. Each seat in the audience area may be either an individual seat for one patron or a coupled seat for two patrons, or a bench for several patrons, or any other type. All of the seats in the audience area may be of the same type or of the different types depending on individual design. The dimensions of the seats may vary substantially depending on the physical parameters (e.g. average body dimensions) of the theater patrons. All of the seats in the audience area may have the same dimensions or different dimensions depending on individual design. For example, the theater may be specifically designed for the kids of different ages. In this case, the dimensions of the seats must be relevant to the average body dimensions of the kids of these ages, living in the area of this theater.

The theater may have any number of screens. Each screen in the theater may be either a conventional projection screen or an LED screen, such as Samsung Onyx LED screen, or any other type. All of the screens in the theater may be of the same type or of the different types depending on individual design. Each screen in the theater may have any dimensions. All of the screens in the theater may have the same dimensions or different dimensions depending on individual design. A resolution of the screen may be any resolution more than "2K" format, which is 2048 x 1080 according to the Digital Cinema System Specification (Version 1.3) published by Digital Cinema Initiatives (DCI), LLC. All of the screens in the theater may have the same resolution or different resolutions depending on individual design. A surface of the screen may be either a flat surface or a curved surface, or any other type. The surfaces of all of the screens in the theater may be of the same type or of the different types depending on individual design. A curvature of the surface of the screen may be allocated along either a horizontal axis or a vertical axis, or both axes (e.g. spherical curvature). All of the screens in the theater may have the same curvatures or different curvatures depending on individual design.

It is to be appreciated that the term ”movie theater” may be also referred to as a moving picture house, a motion picture theater, a picture theater, a film theater, a movie house, a picture house, a film house, a cinema, or any other similar term. The term “theater” may be also referred to as a theatre. The term "plurality," as used herein, means two or more. The term "coupled" means connected to or engaged with, whether directly or indirectly, for example with an intervening member, and does not require the engagement to be fixed or permanent, although it may be fixed or permanent. The terms "first," "second," and so on, as used herein are not meant to be assigned to a particular component so designated, but rather are simply referring to such components in the numerical order as addressed, meaning that a component designated as "first" may later be a "second" such component, depending on the order in which it is referred. It should also be appreciated that designation of "first" and "second" does not necessarily mean that the two components or values so designated are different, meaning, for example, a first direction may be the same as a second direction, with each simply being applicable to different components. The terms “toward,” “forward,” “backward,” “outward,” “rearward,” "upward," "downward," and other orientation descriptors are intended to facilitate the description of the exemplary embodiments of the present invention, and are not intended to limit the structure of the exemplary embodiments of the present invention to any particular position or orientation. Terms of degree, such as "substantially," “relatively,” or "approximately" are appreciated by those of ordinary skill to refer to reasonable ranges outside of the given value, for example, general tolerances associated with manufacturing, assembly, and use of the described embodiments. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly appreciated by one of ordinary skill in the art to which the invention pertains.